

HEDONIC PRICE FUNCTIONS

ROBERT E.B. LUCAS*
University of California, Los Angeles

Three feasible interpretations of cross-sectional hedonic price regression equations are derived from consumer choice, profit maximization by competitive firms, and market clearing, each conceptual experiment deploying a Lancasterian "new approach" to micro theory. Lancaster's theory is compared with those consumer theories of Houthakker heritage, and some limitations of each are indicated. The assertion that Adelman and Griliches' quality-adjusted hedonic price index is a constant satisfaction index is shown to necessitate interpersonal comparisons of utility, and the possibility of identifying demand and supply functions for commodities by exclusion restrictions on characteristics is rejected.

Hedonic price functions are regression equations of the general class:¹

$$(1) \quad \hat{P}_i = P(V_{i1}, \dots, V_{ij}; u_i)$$

where $\hat{P}_i$ = observed price of commodity i ;

V_{ij} = amount of some "intrinsic quality"
(or "characteristic") j per unit of commodity i ;

u_i = a disturbance term.

The purpose of this paper is to provide a theoretical basis for evaluating alternative interpretations and applications of hedonic price functions estimated from cross-sectional data. It is demonstrated that such functions may result from more than one kind of conceptual experiment. First, attention is focussed on the relationships between (1) and consumer choices, and these results are used to reconsider Adelman and Griliches' proposed technique for constructing a consumer price index when the quality of commodities is known to change.² The basic model employed for this analysis is that of Lancaster's "new approach" to consumer

*This is a revised version of Lucas (1972), chapter 2. R. Bailey, R. E. Hall, F. M. Fisher, A. Klevmarken, J. Muellbauer, and S. Rosen contributed many helpful comments.

1. Griliches (1971) lists a number of studies reporting estimates of hedonic price functions.
2. Adelman and Griliches (1961).

theory,³ and two issues are raised with respect to this approach: (i) whether the linear technology of that model is too restrictive for the study of hedonic price functions, (ii) whether the consumer theory of Adelman and Griliches differs substantively from that of Lancaster.

The next type of conceptual experiment considered is a very simple model of profit maximization by perfectly competitive firms, and again it is shown that a plausible hedonic price function can result from such behavior. This second case is consistent with the "cost" interpretation which some investigators have put on (1).⁴

Finally, a third interpretation of hedonic price functions is provided by a model of competitive price determination. The latter draws a link between two branches of empirical work on quality: the estimation of hedonic price functions, and of demand functions which include both prices and characteristics of commodities.⁵ Some limitations on exclusion restrictions for identifying the separate demand and supply functions for commodities complete this final part of the paper.

I. HEDONIC PRICE FUNCTIONS AND CONSUMER CHOICE

Ia. Lancasterian consumer theory and the hedonic price function.

This section briefly summarizes Lancaster's model of consumer theory, and draws a link between an outcome of Lancaster's analysis and estimated hedonic price functions.

Lancaster writes the individual's utility function as,

$$(2) \quad U = U(Z_1, \dots, Z_J)$$

where Z_j is the total amount of characteristic j obtained by the consumer.

A linear consumption technology is assumed to relate the vector of characteristic totals to the quantities of commodities consumed. That is,

$$(3) \quad Z_j = \sum_i V_{ij} q_i \quad j = 1, J$$

where q_i is the quantity of commodity i consumed.

Lancaster assumes that the consumer chooses quantities of the continuously variable commodities such as to maximize utility subject to the consumption technology and the budget constraint. This is written,

3. Lancaster (1966, 1966a, 1968, 1971). For a remarkable, little-known precursor to Lancaster's work, see Gorman (1956).

4. See for example, Fisher, Griliches and Kaysen (1962).

5. One example of this type of study is Quandt and Baumol (1966).

$$\begin{aligned}
 (4) \quad & \text{Max. } U(Z) \\
 & \text{S.T. } Z = Vq \\
 & Y \geq Pq \\
 & Z, q \geq 0
 \end{aligned}$$

where Z is the vector $[Z_j]$ $j = 1, J$
 V is the matrix $[V_{ij}]$ $i = 1, I \quad j = 1, J$
 Y is the consumer's income
 P is a vector of commodity prices $[P_i]$ $i = 1, I$
 q is the vector $[q_i]$ $i = 1, I$

This non-linear program has a solution for the optimal bundle of characteristics, which will be written here as Z^* . Lancaster then considers the most efficient way of obtaining any given bundle of characteristics, such as Z^* . This, he notes, is given by the solution to the program,

$$\begin{aligned}
 (5) \quad & \text{Min. } Pq \\
 & \text{S.T. } Vq \geq Z^* \\
 & q \geq 0
 \end{aligned}$$

The dual of this program is written,

$$\begin{aligned}
 (6) \quad & \text{Max. } \rho Z^* \\
 & \text{S.T. } \rho V \leq P
 \end{aligned}$$

where ρ are the shadow prices of the characteristics.

For those constraints which are binding in the solution of (6),

$$(7) \quad P^a = \rho V^a$$

where P^a is the solution sub-vector of P ;

V^a is the solution sub-matrix of V .

This result is a linear specification of the class of functions (1), the hedonic price function, but how does (7) relate to the *estimated* hedonic price function?

The answer to this question may be developed in four steps:

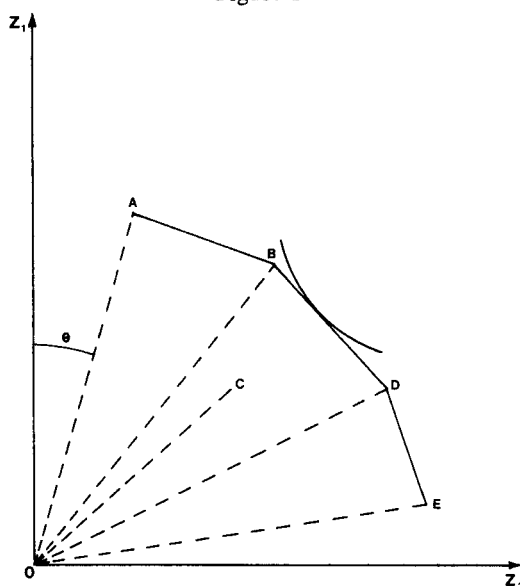
i) First, take the case of a single, representative consumer. Estimates of hedonic price functions are typically obtained by treating each commodity (brand) as an observation. Thus, if one is to estimate the hedonic

function by multiple regression analysis, to have a positive number of degrees of freedom to give confidence in the estimates, it is necessary that the true relation should hold for a greater number of commodities than characteristics.

In the instance of a non-degenerate program, it is well known that the consumer will choose to consume no more commodities than there are characteristics. It follows that (7) holds for at most J commodities and cannot, in this event, be the hedonic price function estimated by multiple regression analysis.⁶

ii) This, however, is not the end of the story for the representative consumer.⁷ Figure 1 illustrates the non-degenerate programming problem, using the familiar Lancasterian diagram, for the case of two characteristics (Z_1 and Z_2) and five commodities (A through E).⁸

Figure 1



6. This rules out gaining degrees of freedom by omitting characteristics from the specification of the estimated form.

7. I am grateful to W. M. Gorman and J. Muellbauer for a discussion of this point. See Gorman (1956) and Muellbauer (1972).

8. A few words of clarification may help the reader who is unfamiliar with this diagram. Commodity A has characteristics Z_1 and Z_2 in the proportion represented by the angle of the ray OA . Thus, $\tan \theta = V_{A2}/V_{A1}$. If the consumer spends all of his income on commodity A , at the given price, he can obtain a combination of characteristics equal to the coordinates of point A . Points B through E are corresponding points for the other commodities.

The locus $ABDE$ is the consumption possibility frontier in characteristic space, as defined by the consumption technology, parametric prices, and budget constraint for the consumer. As drawn, the consumer will choose that combination of commodities B and D which provides characteristics represented by point X . The equalities (7) then hold for commodities B and D , which raises the interesting issue of what happens to the other commodities.

Since there is zero demand for commodities A , C , and E at the going price, they are clearly in excess supply if they appear at all. If the markets are stable, then prices of these commodities fall. The supply of these commodities may fall to zero before they enter the solution set, otherwise prices will continue to adjust until points A through E lie in the same plane. In this degenerate case, the number of commodities with positive consumption levels is indeterminate, but the consumer *may* choose to consume more than J commodities.

Thus, in the case of a representative consumer, when prices have adjusted to the degenerate case, the number of binding constraints in (7) may be greater than the number of characteristics. It is then true that (7) *can*, in general, be estimated by multiple regression analysis, but is it the equation *actually* estimated by most investigators? The degenerate case implies a strictly linear specification of the hedonic price function, yet, as Griliches (1971) notes, "Most of the investigators settle after some experimentation for a semi-logarithmic relationship between prices and characteristics . . ." ⁹

To summarize, if there is a single, representative consumer, then prices in the non-degenerate program are unstable and the program will tend to the degenerate. In the latter case, the equations (7) may hold for a sufficient number of commodities such that one could, in principle, estimate this linear hedonic price function. But, since most investigators reject the linear specification, this does not seem to provide a very reasonable interpretation of the fitted curves.

iii) Consider again the non-degenerate program. For each facet of the consumption possibility frontier ($ABDE$ in Figure 1), a linear hedonic price function (7) holds for the same number of commodities as characteristics. However, such equations hold with a different vector of values of ρ for each facet. Therefore, a piecewise linear function relates commodity prices to characteristics for all commodities on the consumption possibility frontier. This function cannot be estimated by taking break-points in the characteristic variables at values corresponding to vertices in the true

9. Griliches (1971), p. 5.

equation and applying a piecewise linear estimator, since each segment of the piecewise linear equation would again only involve the same number of commodities as characteristics, thus leaving zero degrees of freedom.¹⁰ However, a non-linear function can, in general, be estimated to approximate the surface of flat facets, and indeed this presents us with a plausible interpretation of the estimated non-linear hedonic price functions.

iv) A non-linear hedonic price function capable of estimation by multiple regression analysis, and interpreted as representing variations in commodity prices and characteristics around the consumption possibility frontier, requires that the Lancasterian consumer problem be non-degenerate. If some quantity of every commodity used to estimate the hedonic price function is observed to be consumed, then from *(i)* we know that the single, representative consumer cannot be on the consumption possibility frontier. Thus, the representative consumer must be abandoned.

Any number of consumers facing an identical choice set of commodities, with a common price vector, and constrained by the same consumption technology, has the same shaped consumption possibility frontier in characteristics space.¹¹ Consumers having different preferences, or different incomes and non-homothetic indifference maps, will, in general, be in equilibrium at different points on the "common" consumption possibility frontier. For example, in Figure 2, one consumer may be in equilibrium at *Y* and consume some combination of commodities *A* and *B*, while another consumer chooses to combine *B* and *D* to reach point *X*.

In short, our first conceptual experiment—that of consumers efficiently selecting commodities with parametric prices and characteristics—does give rise to a function:

$$(8) \quad P_i^* = P^*(V_{i1}, \dots, V_{iJ})$$

where P_i^* is the demand reservation price for commodity *i*, and this function can be non-linear if the representative consumer myth is not adhered to.

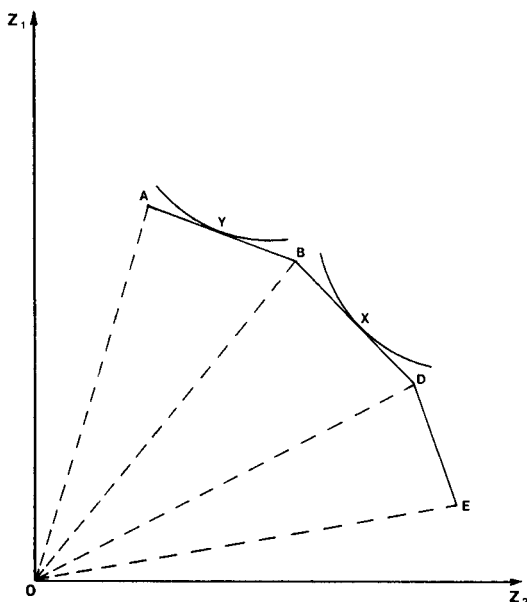
Ib. Hedonic price indices.

One of the chief purposes for which hedonic price functions are intended is that of "correcting" consumer price indices for changes in the quality of the "constant" basket of commodities. The correction technique advocated by Adelman and Griliches (1961), assumes that observed

10. An exception to this is the case when the consumption possibility frontier comprises a series of degenerate segments. For an application of this technique, see Alexander (1973).

11. This frontier is homogeneous of degree one with respect to income.

Figure 2



price changes can be decomposed into two distinct parts, one representing price movements due solely to quality changes, the residual element being called the “poly-genetic” price change. In essence, Adelman and Griliches’ proposal amounts to replacing commodity price changes by their poly-genetic components alone in constructing price indices. To make this technique operational, the authors suggest that the poly-genetic price change, dP_i' , may be defined by,

$$(9) \quad dP_i' \equiv dP_i - \sum_j \frac{\partial \hat{P}_i}{\partial V_{ij}} \cdot dV_{ij}$$

where $\frac{\partial \hat{P}_i}{\partial V_{ij}}$ are the first derivatives of an estimated hedonic price function.

To demonstrate that the corrected, or hedonic, price index represents a constant satisfaction index, Adelman and Griliches consider a model which appears to be quite different from that of Lancaster. First, we shall argue that this apparent difference is largely superficial, and secondly that two difficulties thwart the deduction that hedonic indices represent constant satisfaction indices.

i) The model for Adelman and Griliches may be set out briefly as follows: utility for the individual consumer is taken to be a function of

the quantity of each commodity consumed and the characteristics of each commodity,

$$(10) \quad U = U(q_1, \dots, q_I; V_{11}, \dots, V_{IJ})$$

Consumers choose the q_i and V_{ij} , both of which are assumed to be continuously variable, such as to maximize utility subject to two sets of constraints: the budget constraint and hedonic price functions. The Lagrangean expression may then be written,

$$(11) \quad L = U(q, \underline{V}) - \lambda [Y - \sum_i P_i(V_{i1}, \dots, V_{iJ}) q_i]$$

where $\underline{V}$ is the vector $(V_{11}, \dots, V_{IJ})$ and $P_i(\quad)$ is the hedonic price function for commodity i .

Setting the partial derivatives with respect to the V_{ij} equal to zero and taking ratios gives,¹²

$$(12) \quad U_{V_{rm}} / U_{V_{sn}} = q_r (\partial \hat{P}_r / \partial V_{rm}) / q_s (\partial \hat{P}_s / \partial V_{sn})$$

where $U_{V_{ij}}$ is the marginal utility derived from V_{ij} .

12. Adelman and Griliches include a further assumption, which is both unnecessary and confusing, namely that,

$$(12:1) \quad \partial q_i / \partial V_{ij} = 0$$

The explanation given is as follows:

This latter postulate implies that a typical consumer feels that there is no *direct* substitution between quantity and quality. . . . This assumption does not imply that the quantity of an item purchased is independent of its quality. On the contrary, since quality has a price tag attached to it the quantities of each item consumed depend explicitly on its quality. . .

(Adelman and Griliches (1961) p. 546)

Both the nature and usage of this condition are unclear. No explicit function relating q_i and V_{ij} is given in the model, so what has been differentiated to obtain (12:1)? The last two sentences of the authors' explanation, quoted above, eliminate the possibility that they are referring to equilibrium conditions. The first part of the explanation suggests that they have in mind some constraint on the marginal rate of substitution between q_i and V_{ij} . The latter, however, would be written,

$$(12:2) \quad U_{q_i} / U_{V_{ij}} = (dq_i / dV_{ij}) dU = 0$$

where U_{q_i} is the marginal utility derived from q_i , and not as the partial derivative (12:1).

If it is indeed the right-hand side of the equation (12:2) which Adelman and Griliches wish to imply by (12:1), then one may wonder how they intend to use this constraint, a fact which is not mentioned in the article. Certainly, no such term as $(dq_i / dV_{ij}) dU = 0$ will appear upon differentiation of the Lagrangean expression with respect to the V_{ij} .

In other words, each of the terms $\partial \hat{P}_i / \partial V_{ij}$ used to deflate the consumer price index is proportional to the marginal utility of the relevant characteristic, and from this condition Adelman and Griliches conclude that the deflated consumer price index is a "constant satisfaction" index.

Adelman and Griliches' model attributes the individual consumer with the power of influencing both price and quality of each commodity, a fact which has led to some confusion over causal interpretations of the hedonic price function.¹³ The confusion is easily rectified, once one realizes that Adelman and Griliches have followed Houthakker (1951) in abandoning the traditional concept of a homogeneous commodity. A "commodity" in the Houthakker/Adelman and Griliches approach includes many "varieties" of that commodity, each "variety" being one Lancasterian "commodity."¹⁴ Here we shall retain the term "commodity" for the Lancasterian concept of a homogeneous good, and refer to Adelman and Griliches' concept as a "class of commodities."

To simplify matters, the issue of classes of commodities within the Lancasterian framework will be shelved until section *Ic* below. The essentials of the two models may be understood more simply by assuming that there is only one class of commodities, thus reducing Adelman and Griliches' utility function to,

$$(13) \quad U(q_I; V_{I1}, \dots, V_{IJ})$$

Adelman and Griliches allow the consumer to choose a vector of values for the characteristics of this class of commodities, and assume that some commodity within the class possesses this combination of characteristics. Implicitly, then, the consumer is choosing a commodity within the class, and the price of this particular commodity is exogenously given to him according to the hedonic price function.

The exogenous nature of the hedonic price function in Adelman and Griliches differs somewhat from Lancaster's model, but this distinction is not substantive. Lancaster's analysis separates two aspects of consumer choice: efficiency, and selection of a particular point on the consumption possibility frontier. One way to view Adelman and Griliches' problem is as the second of these aspects taken in isolation: inefficient commodities

13. For example, Ohta (1971), pp. 45-46, writes,

The first supposition of the traditional hedonic approach, that the market price of the commodity reflects the judgement of the consumer on its quality, is not fulfilled in the actual market. The price is often set by the firm rather than by the consumer, and is related to its production costs.

14. Houthakker's analysis is confined to the case of a single characteristic, but the author's concluding remarks on multiple characteristics anticipate some aspects of Lancaster's later work.

have already been eliminated from the market, leaving a systematic relationship between commodity prices and characteristics, which is exogenous to the individual consumer and determines the shape of his consumption possibility frontier in characteristic space.

Finally, note that Adelman and Griliches' assumption that the V_{ij} are continuously variable implies that a continuum of commodities exists. In this extreme case, the consumption possibility frontier in characteristic space is smooth, yet every point on the frontier is a vertex. It is then apparent that the consumer will always choose to consume only one commodity: a point noted explicitly by Houthakker.¹⁵

In short, Adelman and Griliches' method of introducing commodity characteristics into consumer theory does not warrant a separate treatment in our subsequent analysis, since it may be viewed as a special case of the Lancasterian technique, i.e., the case when there exists a continuum of commodities, thus rendering the efficiency frontier smooth, and reducing the solution set of any one consumer to just one commodity within a class.¹⁶

ii) If the estimated hedonic price function (1) is interpreted as approximating the consumer efficiency frontier (8), then (12) follows fairly simply. First, note that the ratio of two first derivatives of (8), evaluated at a point, is roughly equal to the ratio of the shadow prices of these characteristics at that point. From the Kuhn-Tucker conditions we know that the shadow prices, ρ , are proportional to the marginal utilities of the characteristics, for tangency solutions. Therefore, if a consumer is in equilibrium at a tangency solution, then the ratio of the first derivatives of the hedonic price function at that point is roughly equal to the marginal rate of substitution between characteristics for that consumer.

However, we know from section *Ia* that for non-linear hedonic price functions, (12) is not a unique-valued, equilibrium condition. In conventional consumer theory, since all consumers face identical prices, the opportunity locus is linear. Every consumer therefore has the same marginal rate of substitution between commodities in equilibrium, barring corner solutions. But characteristics are not sold separately at common prices, they are sold in the form of packages, and the opportunity locus in characteristic space is non-linear if the hedonic price function is non-linear. Consequently, each consumer has, in general, a different value of

15. See Houthakker (1951), p. 156. For an alternative interpretation, based on linearity of indifference curves, see Muellbauer (1973).

16. On the other hand, Lancaster's $U(\sum q_i V_{iJ}, \dots, \sum q_i V_{iJ})$ is a special case of (10), even in the event of single commodity choices per class. Gorman (1956) considers two possible extensions of this Lancaster-type utility function.

the marginal rate of substitution between characteristics, even in equilibrium. To achieve these many equilibrium marginal rates of substitution we are obliged to abandon the representative consumer, and it follows that to draw any conclusions about the welfare properties of the deflated price index, using a condition such as (12), then necessitates interpersonal comparisons of utility.

A further difficulty for the hedonic price index is that its definition implicitly assumes that the terms $\partial \hat{P}_i / \partial V_{ij}$ remain unchanged from period to period, which, according to our interpretation, is approximately equivalent to assuming that the shadow prices of characteristics are unchanged. This methodology is explicitly intended for circumstances in which the quality of commodities varies through time. Yet in the Lancasterian model, a change in quality must be treated as the introduction of a new commodity, which generally results in a change in shadow prices, not because of a shift in tastes, but because of an efficiency substitution effect.

Ic. On the generality of Lancaster's model.

This section returns to reflect on the general applicability of the basic premises of the Lancasterian framework. These reflections are organized into two sub-sections: the first deals with the generality of Lancaster's model in the absence of discrete variables, the second broaches the issue of discrete variables.

i) Lancaster's model represents a fairly major break from traditional consumer theory, the novelty of the approach residing in its attention to the causes of consumers' desire for particular commodities. However, this transition is achieved at the cost of two particularly restrictive assumptions: first, that utility is assumed to be independent of the joint distribution of characteristics across commodities, i.e., utility depends only on the Z 's; second, that analysis is confined to the case of a linear consumption technology.

Whether or not the former restriction is reasonably sensible can only be resolved in the context of particular cases, but a more general point can be made with respect to the linear technology. To understand this point, one must consider what is meant by a characteristic. If the analysis is confined to "intrinsic physical characteristics" the case is clear. Twice as much meat yields twice as much protein and twice as much fat, hence the technology must be homogeneous of degree one. Further, the amount of protein contained in an egg is not dependent on the amount of meat consumed, so the technology is additive. However, confusion arises when one considers more abstract characteristics such as nourishment. Our bodies are probably not capable of extracting indefinite amounts of nourishment from inordinately large meals, although in some sense the nourishment is

there. The lesson, one suspects, is that such confusion is avoided by confining the analysis to physical intrinsic characteristics, and indeed these are typically the only ones measurable on a cardinal scale.¹⁷

Nourishment is obtained from fat, protein, vitamins and so on, which essentially requires a second step in the consumption technology. If utility is derived directly from nourishment, and nourishment from food characteristics, then a utility function can be written, substituting the second step in the consumption technology into the expression for utility.

$$(14) \quad U = U(h^1(Z_1, Z_2), h^2(Z_3, Z_4), \dots)$$

where for example, $h_1 = h^1(Z_1, Z_2)$ is the amount of nourishment obtained from protein and fat.

We are now in a position to take up the issue of the treatment of classes of commodities within the Lancasterian framework, which was postponed in section *Ia*. In practice, hedonic relations are not estimated for the set of all commodities, but for some particular class of commodities, such as cars. To justify estimating a hedonic price function for cars alone, it is necessary to assume the choice of a car to be independent of the quantities of commodities chosen in all other classes of commodities. In other words, it is necessary to assume the utility function defined in commodity space to be separable.¹⁸

$$(15) \quad U(h(q_1, \dots, q_m), q_{m+1}, \dots, q_I)$$

where the first m commodities are the set of commodities in the class under consideration.

In (14) we see that it is natural to write the utility function in characteristic space as separable. To obtain the condition of separability in commodity space requires the additional condition that the matrix V be

17. Sargan (1972) notes that the assumption of linearity is probably inappropriate for the empirical work in Lancaster (1971), chapter 10. In this chapter, Lancaster considers such abstract characteristics as "size and comfort of seating" and "quietness and performance," which are, of course, measured on an ordinal scale. Lancaster does not attempt to use these ordinal variables in regression analysis, and his purpose is not that of estimating a hedonic price function.

A sharp distinction should also be drawn between the application of a consumption technology to: (a) the intrinsic quality content of commodities (as above), and (b) the concept of household production (as in Muth (1966) and Becker (1965)). Clearly, the above defense of a linear technology is not intended to apply to (b).

18. Fisher and Shell (1968) comment on separability in a related context. Note that it is not necessary for our argument that any Z_i enters only one h_i function, and that some restrictions must be placed on these h_i functions if the utility function is to remain quasi-concave. For further desirable restrictions on the Lancasterian utility function in the context of constructing Laspeyres-type hedonic price indices, see Muellbauer (1975).

decomposable. In the context of (14) and (15) this means that Z_1 and Z_2 depend only on $q_1, \dots, q_m$. This latter assumption can be justified simply by a suitable choice of characteristic categories. Of course, the consumption possibility frontier in the space of car characteristics must then be defined as an iso-expenditure curve, rather than being subject to constraint by the consumer's income.

ii) In empirical work, one frequently encounters circumstances in which two of the sets of variables in the Lancasterian model can assume only discrete values: i.e., when quantities of commodities are not continuously variable, and when the V_{ij} are dummy variables. In order to distinguish the effects of these two, they will initially be taken separately here, beginning with the case of dummy characteristic variables whilst retaining continuously variable commodity quantities.

In the introduction to this paper V_{ij} is defined as the amount of characteristic j per unit of commodity i . If, for some j ,

$$(16) \quad V_{ij} = 1, 0 \quad \text{all } i,$$

then V_{ij} is clearly not defined in terms of the units of measurement of the commodity. Application of Lancaster's linear technology under these circumstances results in Z_j measuring the sum of the quantities of different commodities which possess characteristic j . Obviously, Z_j is then sensitive to the units of measurement of commodity quantities, and has no clear economic meaning. It must be concluded that Lancaster's framework is not very suitable under these circumstances.¹⁹

So far it has been assumed that the activity levels—the quantities of commodities consumed—are continuously variable. Indivisibilities have long plagued conventional consumer theory and no really satisfactory solution has been found. The characteristics approach probably does no better and no worse than conventional theory in facing indivisibilities, but in empirical work the hedonic relation has been applied almost exclusively to durable goods which are very lumpy by nature, such as cars, refrigerators and houses. Estimates of hedonic price functions are typically

19. Lancaster (1971), pp. 107-8, writes:

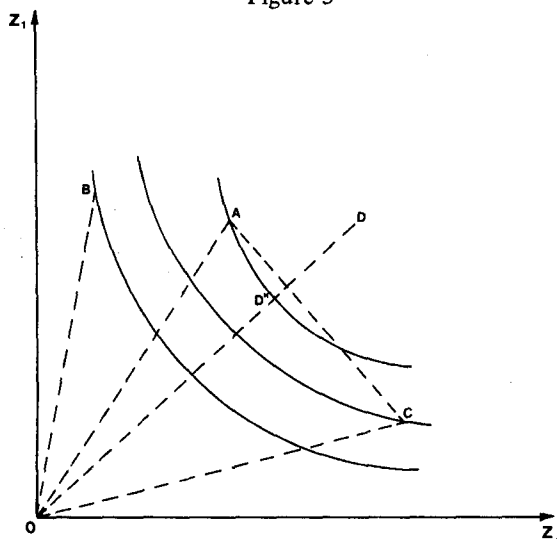
There is, however, one important class of non-linearities which are both prevalent and relate to characteristics properly defined. This is the class of discrete or integer characteristics. Some things, for example, do or do not have a certain characteristic, but it is not meaningful to speak of "more" or "less" of the characteristic.

It is not at all clear what advantage would be gained by admitting a non-linear technology to deal with this issue. What has to be settled is what one would wish Z_j to represent in the instance of dummy characteristics. For example, if $V_{ij} = 1$ when commodity i is white, zero otherwise, what sense of "whiteness" is one really trying to capture by the variable Z_j ?

obtained by treating, for example, one car as the unit of quantity. Such an approach will generally yield a consumption possibility frontier which does not correspond to the exhaustion of the consumer's budget. A more satisfactory approach would be to convert the activity levels and commodity prices into flow variables.

A further difficulty with discrete consumption activities is that the consumption possibility frontier ceases to be continuous, and is comprised only of a set of unconnected points. In the extreme case, when the consumer's budget is such that he is able to choose only one item (such as one car), the consumption possibility frontier consists only of the vertex points. This problem is illustrated in Figure 3.

Figure 3



The consumer is faced with a choice between A , B , and C , and given the indifference map shown he will clearly choose A . This raises some doubt as to whether a consumer in equilibrium would have an indifference curve which is even approximately tangential to a function interpolated through the points of the consumption possibility frontier. The tangency condition must be nearly approximated if the different commodities are dense along the consumption possibility frontier, and there is reason to believe that this will be the case, at least in the neighborhood of commodities chosen by many consumers. Suppose that a producer decides to market a new commodity, (i.e., a new bundle of characteristics), with characteristics combined in proportions represented by the ray OD . If this new good is so priced that its coordinates are greater than D^* , then

the consumer will, clearly, switch to purchasing it.²⁰ Whether D can be so priced is a question of its production costs, but, at least from the demand side, there is some inducement to introduce commodities which are cross-combinations of previously existing bundles. Hence, there may be some tendency for commodities to cluster in the neighborhood of popular choices.²¹

As a final permutation, the two sets of discrete variables may occur simultaneously. In the extreme case of discrete activity levels, when the consumer can choose only one item, the difficulty of the units of measurement in the presence of dummy characteristics disappears. Under these circumstances the consumption technology may be written,

$$(17) \quad \begin{aligned} Z_j &= \sum_i V_{ij} q_i \\ q_i, V_{ij} &= 1, 0 \quad \text{all } i \\ \sum_i q_i &= 1 \end{aligned}$$

It is then apparent that $Z_j = V_{ij}$ for that single commodity chosen by a consumer. For a consumption technology with these features, the Lancasterian framework is again appropriate, and the same approximate relationships between the hedonic price function and consumer utility functions hold as above.

II. HEDONIC PRICE FUNCTIONS AND PRODUCTION COSTS

This second part of the paper contains some brief notes on an alternative interpretation placed on the hedonic price function by some investigators:²² namely, that this function is related to production costs.

Since this analysis is meant only to be illustrative, let us assume a particularly simple form of the total cost function for the firm,

20. Lancaster (1966b) notes the peculiarity of this feature, capturing the market with a commodity which may represent a combination of characteristics within the imaginary boundary CA .

21. The latter qualification suggests the presence of economies of scale in production.

22. See, for example, Fisher, Griliches, and Kaysen (1962). Rosen (1974), written concurrently with this paper, takes a different approach to this issue—that of a Houthakker rather than Lancasterian framework. One consequence of this difference is that it remains unclear as to how the q 's drop out of the firms' hedonic price function in Rosen's paper.

$$(18) \quad C = C(Z; r)$$

$$Z = Vq$$

where r is a vector of factor prices.²³

Any efficient firm faced with given prices, will choose to produce quantities of commodities such as to minimize costs subject to receiving a minimum level of revenue. This can be written,

$$(19) \quad \text{Min.} \quad C(Z; r)$$

$$\text{S.T.} \quad Z = Vq$$

$$Pq \geq R^*$$

$$Z, q \geq 0$$

where R^* is the revenue constraint.²⁴

This non-linear program immediately resembles the problem (4) in consumer theory, and there is no need to repeat the exposition of that model. For a non-linear hedonic price function to represent a cost minimizing solution, firms with either (i) access to different factor prices, or (ii) different cost functions, or (iii) non-homothetic cost functions and different scales of production, must be contained within the industry.²⁵ The efficiency price locus for this case of heterogeneous firms may be written:

$$(20) \quad P_i^{**} = P^{**}(V_{i1}, \dots, V_{iJ})$$

where P_i^{**} is the supply reservation price of commodity i .

The implication is that the estimated hedonic price function may be interpreted such that its derivatives are proportional to the marginal costs

23. Like the Lancasterian utility function, this cost function is particularly restrictive in assuming costs to be independent of the joint distribution of characteristics across commodities. Notice that costs are a function of quantities of commodities produced, and that the linear technology does not necessarily imply that costs are homogeneous of degree one with respect to commodity quantities.

24. The level of revenue constraining this frontier is irrelevant to our purposes, since the frontier is homogeneous of degree one with respect to revenue.

25. Alternatively, the assumption of perfect competition in the product markets must be dropped in favor of markets typified by product differentiation or other forms of imperfect competition.

of characteristics, for any firm which is in a tangency solution at the relevant point on the iso-revenue surface.

III. SUPPLY, DEMAND AND THE HEDONIC PRICE FUNCTION

Although the terms “supply” and “demand” have been mentioned, casually, at some points in the paper, we have not yet considered how such functions may be derived in a “characteristics” framework. This final part of the paper shows how demand and supply functions are approximately related to the hedonic price function, and notes a limitation on identifying information.

Assume that the markets for all commodities are perfectly competitive. Each consumer faces given prices and the analysis of part I is relevant as long as the consumers are in equilibrium (i.e., if the markets are on their demand curves). Each firm also faces infinitely elastic demand curves and the analysis of section II is applicable if all firms are in equilibrium (i.e., if the markets are on the supply curves). If all markets are in equilibrium, then both a cost and a preference approach to the relationship between prices and characteristics are valid. If any markets are out of equilibrium, some a priori information about the operation of the markets in disequilibrium is necessary before any interpretation of “hedonic” price functions can be validated.

In perfectly competitive markets, prices are determined by the interaction of supply and demand. The role of characteristics in determining prices then depends on the influence of these characteristics on the demand and supply functions. For any group of people with tastes such that their indifference curves are tangential to a common facet of the consumption possibility frontier we know that,²⁶

$$(21) \quad P_i^* = k \sum_j U_j \cdot V_{ij}^* \quad \text{for the commodities} \\ \text{purchased by this group,}$$

where $U_j = \partial U / \partial Z_j$, k is some constant.

Assuming that the utility function is such that these expressions can be solved for the non-zero q_i , we can write,

$$(22) \quad q_{iD} = f(p_I, \dots, p_I; V_{II}, \dots, V_{IJ}, \dots, V_{IJ}),$$

given tastes and incomes, and where q_{iD} is the quantity of i demanded by this group.

26. If the commodities are lumpy we know that this is only an approximation.

Summing across all groups on facets adjacent to the vertex of the commodity in question we get,²⁷

$$(23) \quad Q_{iD} = D(p; \underline{V})$$

where Q_{iD} is the market demand for the commodity i .²⁸

Similarly on the supply side we can derive an expression for the market supply function, holding factor prices constant,²⁹

$$(24) \quad Q_{iS} = S(p; \underline{V})$$

If the markets clear, then equating (23) and (24) and solving for prices gives,³⁰

$$(25) \quad P_i^{***} = P^{***}(V_{i1}, \dots, V_{iJ})$$

where P_i^{***} is the market clearing price for commodity i , which provides our third plausible hedonic price function. Notice that if one believes that the estimated hedonic price function is a consequence of the kind of conceptual experiment described in this last section, then $P_i^{***} = P_i^{**} = P_i^*$, and both a "cost" and "utility" interpretation of the results are justified.

If one believes that observed prices do correspond to P_i^{***} , then can one proceed to identify the demand and supply functions by placing a priori exclusion restrictions on the vectors of characteristics in the two functions?

To identify the two functions by means of excluding characteristics requires that at least one characteristic which appears in the demand function can be assumed, a priori, not to affect supply, and vice versa. But, since the observations are all presumed to be equilibria, different vectors of characteristics *cannot* be defined for the supply and demand functions to provide exclusion restrictions for identification. If a characteristic enters the utility function, but is costless, then, in equilibrium,

27. For groups in equilibrium actually at the vertex, no equality (21) holds. In fact, their equilibrium can only be described by inequalities. This problem is ignored in what follows.

28. Note that a change in price will, in general, lead to an efficiency substitution effect which may involve new population groups in the purchase of the commodity.

29. The functions (23) and (24) are common to all commodities since the matrix V is assumed to describe all relevant aspects of the various commodities. For specific examples of the derivations of these market demand and supply functions, see Rosen (1974).

30. Or, more simply equating (8) and (20) and solving for $P_i^{***} = P_i^{**} = P_i^*$.

commodity prices will be invariant to the level of this characteristic. Similarly, if it is costly to add some feature to a commodity but the consumer gains no utility from it then in equilibrium the commodities including this feature will disappear from the market. Consumers and producers may describe their preferences and costs in terms of different characteristics, but in equilibrium this must either be because the two are proxies for the same feature, or because there exists a high degree of collinearity between the characteristics. In either case, exclusion of characteristics on such a basis cannot be used to identify the separate functions. This, of course, does not deny that the demand and supply functions can be identified, merely that this cannot be achieved by exclusion restrictions upon characteristics. A priori information about the *way* in which costs and demand are influenced by characteristics may be used for identification, or the exclusion of income from supply functions and factor prices from demand functions may be used.³¹

Concluding Remarks

The analysis of this paper has been restricted to the realm of competitive markets, no consideration having been given to the more complex subject of monopolistic competition. Within the context of our competitive model in which all actors are price and quality takers, three distinct conceptual experiments have been shown to provide hedonic price equations: efficient commodity choice by consumers and by firms, and market clearing. In the first of these, the locus of commodity prices and qualities around the Lancasterian consumption possibility frontier provides a hedonic price function. However, an experiment in terms of a representative consumer only provides a function with positive degrees of freedom if the choice program is degenerate. The application of regression analysis to hedonic equations, combined with a preference for non-linear specification, must therefore lead us to reject this conceptual experiment as a satisfactory interpretation of observed hedonic relationships. But once consumers with different tastes, or different incomes and non-homothetic indifference maps, are admitted to the story, the resultant non-linear price-qualities locus does provide an acceptable explanation. The obligation to abandon the representative consumer implies that any welfare statements concerning Adelman and Griliches' proposed quality-adjusted price index necessitates interpersonal comparisons of utility. A further complication for this index arises from the instability of the hedonic equation as new commodities are introduced, thus rendering Adelman

31. Rosen (1974) notes some of the latter possibilities, but omits discussion of identification by exclusion restrictions on characteristics.

and Griliches' definition of the poly-genetic price changes ambiguous, depending on whether base or current period cross-sectional hedonic equations are employed.

Interpreting hedonic regression equations as estimates of the Lancasterian consumer efficiency frontier presumes that this "new approach" is a suitable framework. But Lancaster's model is particularly restrictive in ignoring the effect of the dispersion of characteristics across commodities upon utility, and in assuming a linear technology. In pondering the sufficiency of the latter, it is important to distinguish two broad categories of qualities—objective and abstract—the dividing line adopted here being measurement on an objective, cardinal scale. Thus, weight of an automobile is an objective characteristic, whilst comfort is an abstract quality. A linear technology is quite adequate for objective, continuous characteristics whereas it may not be for abstract qualities. If utility is "really" derived from abstract qualities, and these in turn are acquired from the objective characteristics (thus comfort of a car is a function of its weight, length, etc.), then the Lancasterian utility function is separable in objective characteristic space. If the objective characteristic variables are defined such as to provide a decomposable linear technology matrix, then the utility function is also separable in commodity space, and the hedonic price function for any given class of commodities is not affected by the qualities and quantities of every other class of commodity.

Dummy variable characteristics cannot be conveniently handled by the Lancasterian model, except in the case that the solution set of commodities is confined to one commodity type. Nor can this problem be rectified by simply generalizing to a non-linear technology, for the heart of the problem is knowing what meaning one would wish to attach to the characteristic totals, Z , in this context.

The weakest aspect of Lancaster's consumer theory is probably the utility function, in that utility depends only on the characteristic totals, Z , and not on their joint distribution across commodities. The Houthakker, Adelman-Griliches, Rosen, alternative approach does deploy a more general utility function, but this generality is achieved at the cost of assuming a continuum of commodities within each class, a continuum of consumers, and each consumer choosing only one commodity within a class. This more general utility function cannot be readily applied to Lancaster's model, for if one is dealing with Lancaster-cum-traditional homogeneous goods, then the quality variables in (10) are rendered superfluous.

The second conceptual experiment in the paper is the firm's efficient selection of commodity packages to be produced. If a competitive firm minimizes costs, which are taken to be a function of characteristics totals and factor prices, subject to a linear commodity-characteristic technology

and a revenue constraint, a hedonic price function with first derivatives reflecting characteristic costs follows, with all the qualifications and reservations of the first conceptual experiment.

Finally, once it is realized that "utility" and "cost" hedonic price functions are really no more than inverted market demand and supply functions respectively, it follows that these two interpretations of the same function are compatible, if markets clear. In this event, exclusion restrictions on characteristics cannot be used to identify the separate demand and supply functions.

REFERENCES

- Adelman, I. and Griliches, Z., "On an Index of Quality Change," *American Statistical Association Journal*, Sept. 1961, 535-548.
- Alexander, W. E., "The Hedonic Theory and Measurement of Quality Change," paper presented to the Canadian Economics Association meetings, Kingston, 1973.
- Becker, G. S., "A Theory of the Allocation of Time," *Economic Journal*, Sept. 1965, 75, 493-517.
- Court, A. T., "Hedonic Price Indexes with Automotive Examples," *The Dynamics of Automobile Demand*, 1939.
- Fisher, F. M., Griliches, Z., and Kaysen, C., "The Cost of Automobile Model Changes Since 1949," *Journal of Political Economy*, Oct. 1962, 70, 433-51.
- Fisher, F. M. and Shell, K., "Taste and Quality Change in the Pure Theory of the True Cost-of-Living Index," *Value, Capital and Growth: Essays in Honour of Sir John Hicks*, Ed. J. N. Wolfe, Edinburgh, 1968.
- Gorman, W. M., "A Possible Procedure of Analyzing the Quality Differentials in the Egg Market," (mimeo), 1956.
- Griliches, Z., "Hedonic Price Indexes for Automobiles: An Econometric Analysis of Quality Change," *The Price Statistics of the Federal Government*, General Series, No. 73, New York: National Bureau of Economic Research, 1961.
- , "Introduction: Hedonic Prices Revisited," *Price Indexes and Quality Change*, Ed. Z. Griliches. Cambridge, Mass., 1971.
- Houthakker, H. S., "Compensated Changes in Quantities and Qualities Consumed," *Review of Economic Studies*, 1951-52, 19, 155-164.
- Lancaster, K. J., "New Approach to Consumer Theory," *Journal of Political Economy*, April 1966, 74, 132-57.
- , "Change and Innovation in the Technology of Consumption," *American Economic Review Papers and Proceedings*, May 1966, 56, 14-28, (b).
- , *Mathematical Economics*, New York, 1968.
- , *Consumer Demand: A New Approach*, New York, 1971.
- Lucas, R. E. B., "Working Conditions, Wage-Rates and Human Capital: A Hedonic Study," Ph.D. Dissertation, M.I.T., 1972.

Muellbauer, J., "Review of Griliches (1971)," *Journal of Economic Literature*, 1972, 10, 830-32.

_____, "The Divisia Index Approach to Quality Change: A Note on Adelman and Griliches," Birkbeck College Discussion Paper No. 9, London, 1973.

_____, "Household Production Theory, Quality and the 'Hedonic Technique,'" *American Economic Review*, forthcoming, March 1975.

Muth, R. F., "Household Production and Consumer Demand Functions," *Econometrica*, July 1966, 34, 699-708.

Ohta, M., "Hedonic Price Index for Boiler and Turbo-Generator: A Cost Function Approach," Technical Report No. 40, *Project for the Evaluation and Optimization of Economic Growth*, Institute of International Studies, Berkeley, June 1971.

Quandt, R. E. and Baumol, W. J., "The Demand for Abstract Transport Modes: Theory and Measurement," *Journal of Regional Science*, Winter 1966, 6, 13-26.

Rosen, S., "Hedonic Prices and Implicit Markets," *Journal of Political Economy*, 1974, 82, 34-55.

Sargen, J. D., "Review of Lancaster (1971)," *Economic Journal*, December 1972, 82, 1416-17.